

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

**Duration : 1 Hour
Total Marks:50**

Annexure A

Analytical Problem Solving

Code of Conduct:

I M. Pavankalyan (192312178) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

*Level - 3
50*

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Sub: AI
Co1 - AT1
code: CSA17

① Online Search Agent in an unknown Building:

An online Search Agent explores an environment while acting, without having a complete map beforehand. It learns about the environment as it moves.

Data exploration:

1. The robot starts at an unknown location.
2. It senses only its immediate surroundings.
3. It selects a safe unexplored direction and vision.
4. Cleaning robot is placed at an unknown location in building.
5. No map of the building is available.

Goal Site:

Explore the entire building and clean all reachable areas.

Path cost:

Cost is measured by the total distance travelled, time taken, or energy consumed.

Challenges compared to offline search

- No complete map is available initially.
- Decisions are made based on local information only.
- Frequent backtracking may occur.
- More time and energy are required.

② Smart thermostat Agent Design:

Simple reflex agent:

- Current room temperature.
- Target temperature.

Characteristics:

- Makes decisions based only on the current temperature.
- Does not remember previous temperature or systems status.
- Suitable for simple environments.

Model based reflex agent:

- Current room temperature.
- Target temperature.
- Estimated room conditions.
- Update the internal state after each reading.

important cases. The student
can use a copy of guide
and transmitter logs
light.

Whole
ing and cycling
and Cape location
actions

to make
left right

Internal state:

- Previous temperature readings.
- Current status of heater A/C.
- Rate of temperature change.
- Estimated room conditions.

Rules:

- Update the internal state after each reading.
- If temperature is below the target and still with the A.C.
- If target temperature is reached

③ Self-driving car on city roads:

- Road lanes and traffic sign.
- Traffic lights.
- Nearby vehicles.
- Pedestrians and cyclists.
- Speed and GPS location.

Possible actions:

- Accelerate.
- Brake.
- Steer left/right.

Operating Environment:

Sensors cannot always detect everything due to blinds.

Determinism: other vehicles and pedestrians behave unpredictably.

Nature: Dynamic - The environment changes continuously while the car is moving.

State Space: Continuous: Position, Speed, Steering angle, and time change continuously.

Suitable agent architecture:

considers multiple factors such as safety, travel time, passenger comfort, and fuel efficiency.

- Selects the action that maximizes overall utility.

- Handles uncertain and dynamic environment effectively.

- Makes better decisions of the based agent.